

Cloud Computing Experiment No 1

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Class: TE Comp A (Batch B)

1. Aim

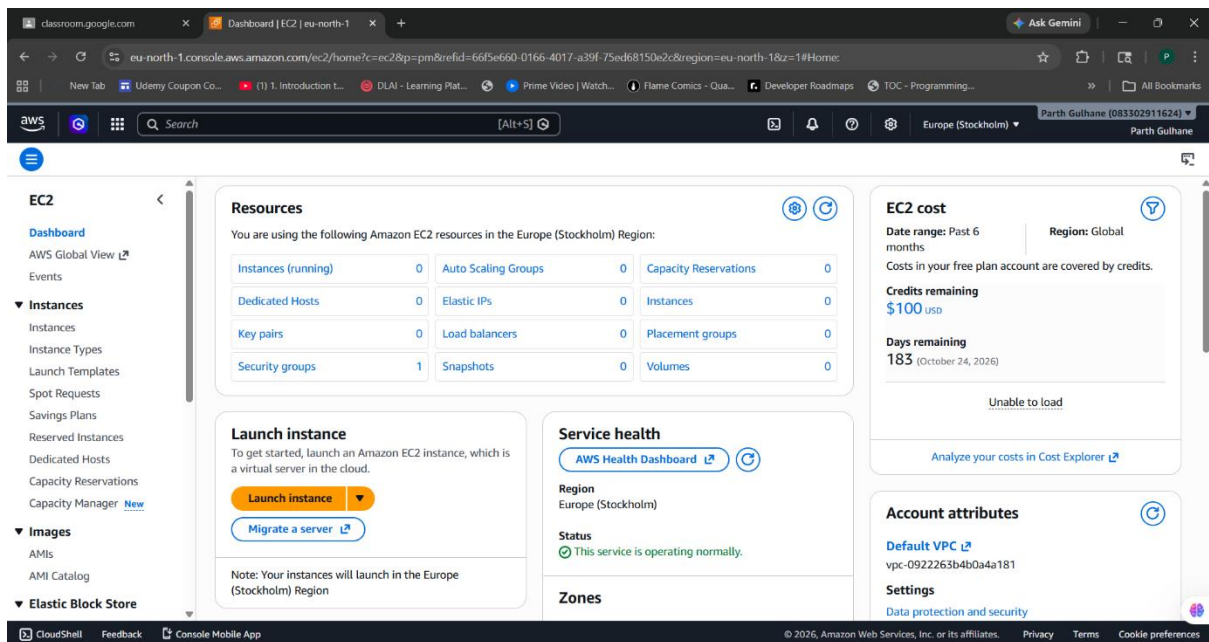
To launch an EC2 instance on AWS and host a static website using Apache web server.

2. Objective

To understand cloud computing services and deploy a simple static website on AWS EC2 instance.

3. Implementation With Output

Step 1: Open AWS EC2 Dashboard



Explanation: The AWS EC2 dashboard is opened to manage cloud resources such as instances, volumes, and security groups

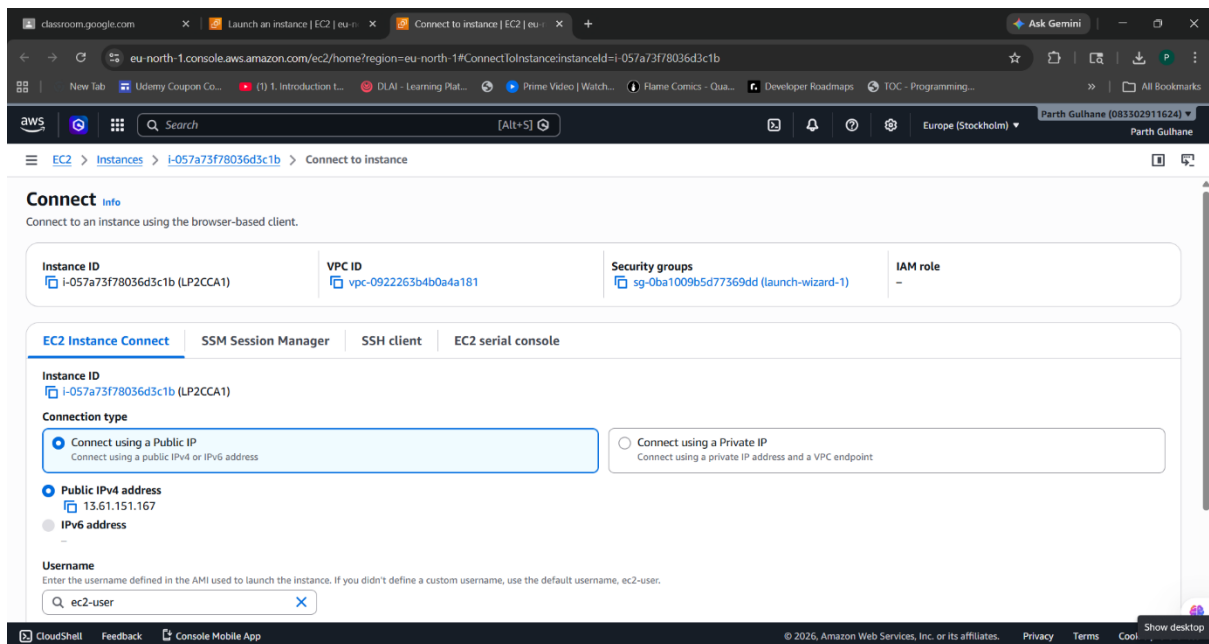
Step 2: Launch an EC2 Instance

- Navigate to EC2 → Instances → Launch Instance
- Choose AMI: Amazon Linux 2023
- Instance Type: t2.micro (Free tier eligible)

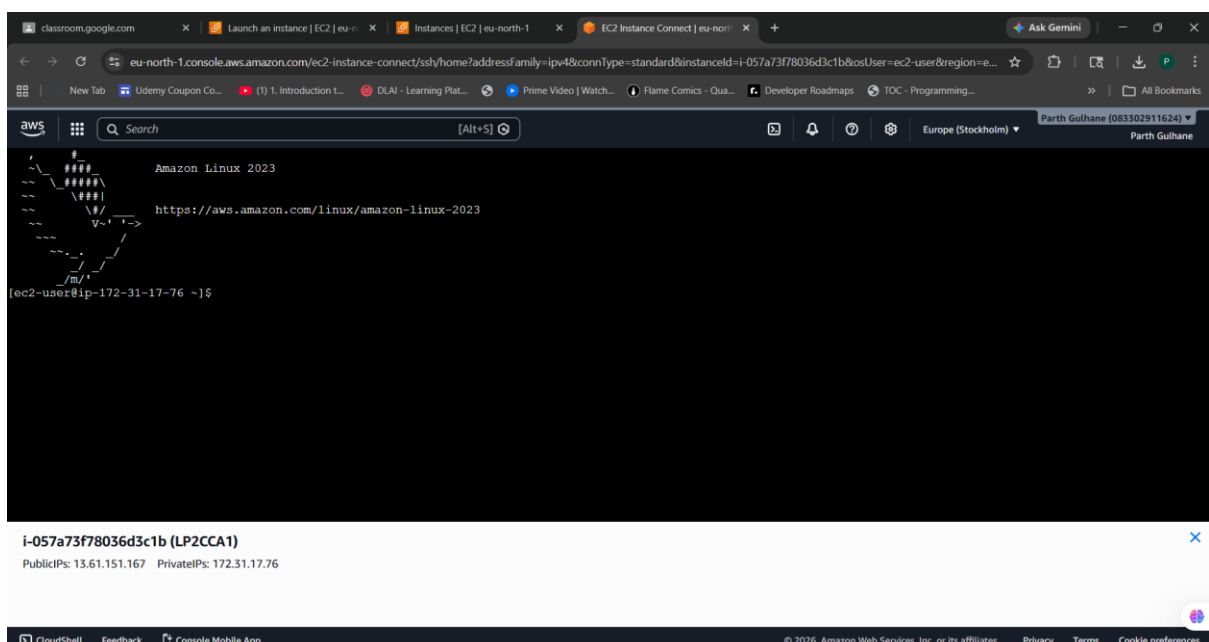
- Configure Security Group: Allow SSH (port 22) and HTTP (port 80)
- Launch and download the .pem key pair
- Note the Public IP address assigned to the instance

Step 3: Connect To Instance

Use the AWS Console's built-in 'Connect' feature (EC2 Instance Connect) to open a browser-based terminal session:



Step 4: Click on Connect



Step 5: install Apache Server

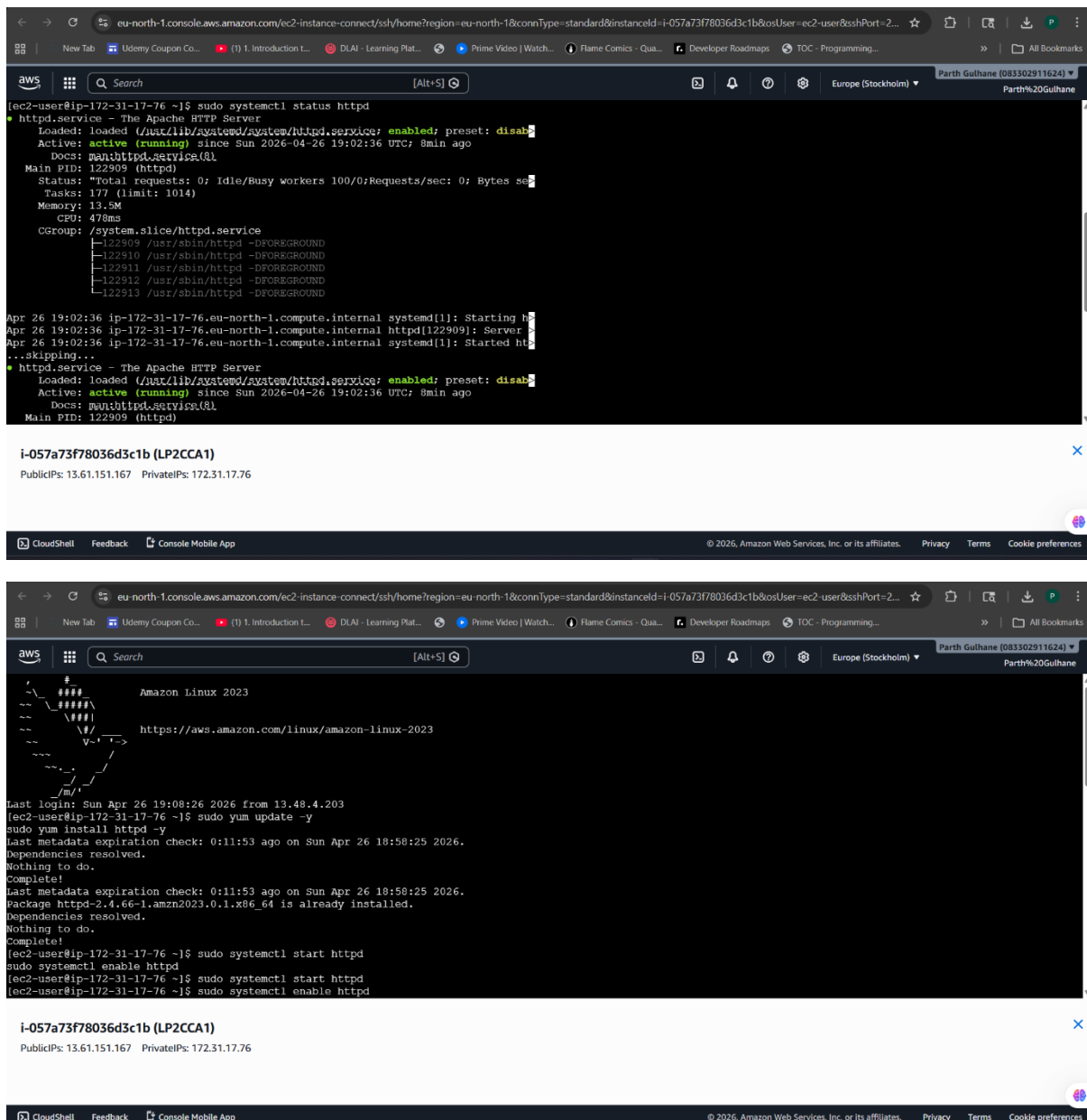
Explanation:

Apache HTTP server is installed using the command:

- `sudo yum install httpd -y`

The installation process downloads and installs required packages successfully.

Step 6: Start and Enable Apache Server



```
ec2-user@ip-172-31-17-76 ~$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-04-26 19:02:36 UTC; 8min ago
     Docs: man:httpd.service(8)
   Main PID: 122909 (httpd)
  Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served: 0"
    Tasks: 177 (limit: 1014)
   Memory: 13.5M
      CPU: 478ms
   CGroup: /system.slice/httpd.service
           └─122909 /usr/sbin/httpd -DFOREGROUND
             └─122910 /usr/sbin/httpd -DFOREGROUND
               └─122911 /usr/sbin/httpd -DFOREGROUND
                 └─122912 /usr/sbin/httpd -DFOREGROUND
                   └─122913 /usr/sbin/httpd -DFOREGROUND

Apr 26 19:02:36 ip-172-31-17-76.eu-north-1.compute.internal systemd[1]: Starting httpd.service: The Apache HTTP Server.
Apr 26 19:02:36 ip-172-31-17-76.eu-north-1.compute.internal httpd[122909]: Server built for Linux
Apr 26 19:02:36 ip-172-31-17-76.eu-north-1.compute.internal systemd[1]: Started httpd.service: The Apache HTTP Server.
...skipping...
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-04-26 19:02:36 UTC; 8min ago
     Docs: man:httpd.service(8)
   Main PID: 122909 (httpd)
```

i-057a73f78036d3c1b (LP2CCA1)

PublicIPs: 13.61.151.167 PrivateIPs: 172.31.17.76

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Sun Apr 26 19:08:26 2026 from 13.48.4.203
ec2-user@ip-172-31-17-76 ~$ sudo yum update -y
sudo yum install httpd -y
last metadata expiration check: 0:11:53 ago on Sun Apr 26 18:58:25 2026.
Dependencies resolved.
Nothing to do.
Complete!
last metadata expiration check: 0:11:53 ago on Sun Apr 26 18:58:25 2026.
Package httpd-2.4.66-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
ec2-user@ip-172-31-17-76 ~$ sudo systemctl start httpd
sudo systemctl enable httpd
ec2-user@ip-172-31-17-76 ~$ sudo systemctl start httpd
ec2-user@ip-172-31-17-76 ~$ sudo systemctl enable httpd
```

i-057a73f78036d3c1b (LP2CCA1)

PublicIPs: 13.61.151.167 PrivateIPs: 172.31.17.76

Explanation:

The server is started and enabled using:

- sudo systemctl start httpd
- sudo systemctl enable httpd

The status shows active (running), confirming successful execution.

Step 7: Create HTML Webpage

The first screenshot shows the terminal output of the `systemctl status httpd` command, confirming that the httpd service is active (running). It also shows the command `cd /var/www/html` and the creation of an `index.html` file using `nano`.

```

Main PID: 122909 (httpd)
Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes se
Tasks: 177 (limit: 1014)
Memory: 13.5M
CPU: 478ms
CGroup: /system.slice/httpd.service
├─122909 /usr/sbin/httpd -DFOREGROUND
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Apr 26 19:02:36 ip-172-31-17-76.eu-north-1.compute.internal systemd[1]: Starting h
Apr 26 19:02:36 ip-172-31-17-76.eu-north-1.compute.internal httpd[122909]: Server
Apr 26 19:02:36 ip-172-31-17-76.eu-north-1.compute.internal systemd[1]: Started ht

[ec2-user@ip-172-31-17-76 ~]$ cd /var/www/html
[ec2-user@ip-172-31-17-76 html]$ sudo nano index.html
[ec2-user@ip-172-31-17-76 html]$ sudo chmod -R 755 /var/www/html
  
```

The second screenshot shows the content of the `index.html` file created in the `/var/www/html` directory. The file contains a simple HTML structure with a title and a welcome message.

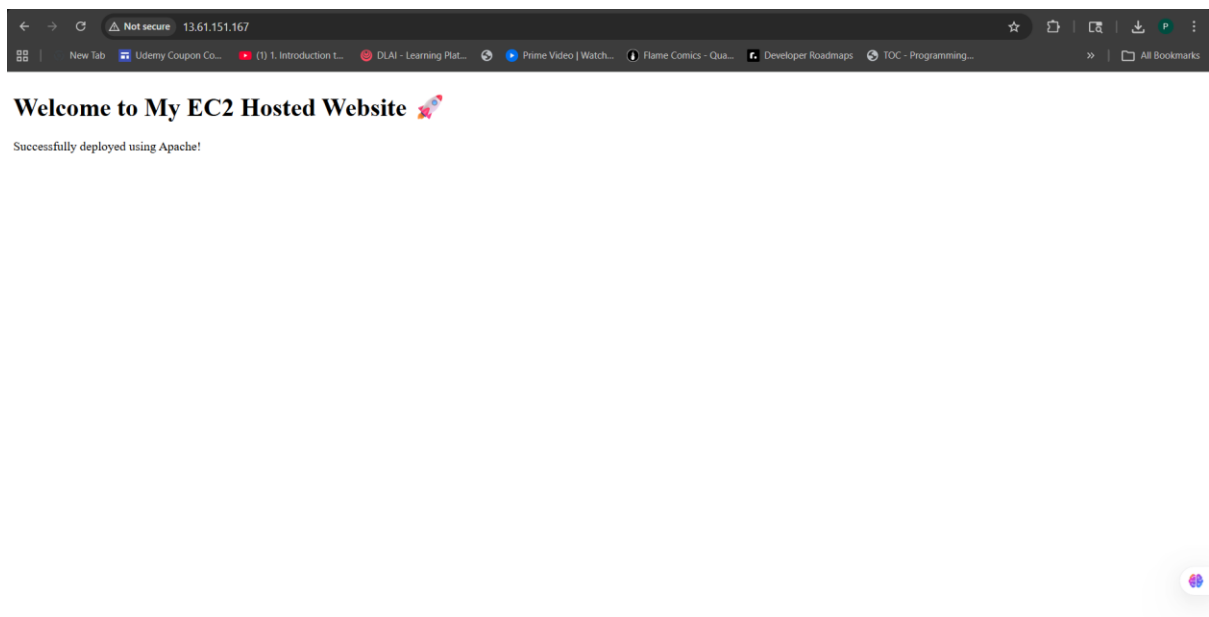
```

GNU nano 2.9.3 index.html Modified
<!DOCTYPE html>
<html>
<head>
  <title>My AWS Website</title>
</head>
<body>
  <h1>Welcome to My EC2 Hosted Website 🚀</h1>
  <p>Successfully deployed using Apache!</p>
</body>
</html>
  
```

Explanation:

Open GNU nano and create an `index.html` page

Step 8: Access Hosted Website



Explanation:

The website is accessed using the public IP address in a browser. The output confirms that the static website is successfully hosted.

4. Conclusion

The static website is successfully deployed and accessed using AWS EC2.